

Telephone Tapping Using FM

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This is an interesting circuit which you can connect to a telephone line to listen to the conversations through any FM radio receiver. Power for the circuit is drawn from the telephone line itself, so it does not require a power supply of its own. The telephone wires act as the antenna for the audio signal.

Circuit and working

Fig. 1 shows the circuit diagram of telephone tapping circuit using FM. The circuit is built around transistors T1 and T2 and bridge diodes D1 through D4. Coils L1 through L3 are made of

24SWG enamelled copper wire wound over 5mm air core. L1 is made of 8 turns, L2 of 6 turns and L3 of 10 turns with a tapping at fourth turn. Variable capacitor VC1 is used for tuning the transmitted frequency. Diodes D1 through D4 rectify the telephone line voltage which drives the circuit.

Working of the circuit is simple. Connect the circuit in parallel to the telephone line to listen to the conversation between the calling and the called parties. Keep switch S1 on and tune your FM radio until you hear the conversation.

Construction and testing

An actual-size, single-side PCB for



PARTS LIST

Semiconductors:

- D1-D4 - 1N4148 diode
- T1, T2 - BC547 npn transistor

Resistors (all 1/4-watt, 5% carbon):

- R1 - 220-ohm
- R2 - 33-kilo-ohm
- R3 - 10-kilo-ohm
- R4 - 47-kilo-ohm
- R5 - 390-ohm

Capacitors:

- C1, C3 - 22pF ceramic disk
- C2 - 100nF ceramic disk
- C4 - 1nF ceramic disk
- C5 - 10pF ceramic disk
- C6, C7 - 47pF ceramic disk
- C8 - 100µF, 25V electrolytic
- VC1 - 50pF variable capacitor

Miscellaneous:

- CON1 - 2-pin connector
- S1 - On/off switch
- L1 - 8 turns, 5mm dia, 24SWG air core
- L2 - 6 turns, 5mm dia, 24SWG air core
- L3 - 6 turns + 4 turns, 5mm dia, 24SWG air core

Test Points

Test point	Details
Across TP0 and TP1	9V-12V, provided S1 is switched on during conversations
Across TP2 and TP3	6V (approx.)

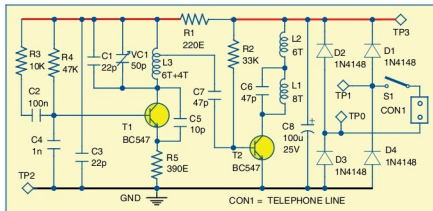


Fig. 1: The telephone tapping circuit

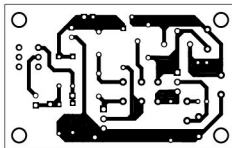


Fig. 2: An actual-size PCB layout for the circuit

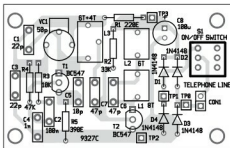


Fig. 3: Component layout for the PCB

tapping a telephone line using FM is shown in Fig. 2 and its component layout in Fig. 3. After assembling the circuit on PCB, enclose it in a suitable plastic box.

Fix 2-pin connector CON1 for connecting the telephone line. Fix switch S1 on front panel of the box. Before using the circuit, verify the test point voltages mentioned in the table. ●

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